

**MODEL TEST PAPER 9**  
**FINAL COURSE: GROUP – I**  
**PAPER – 2: ADVANCED FINANCIAL MANAGEMENT**  
**ANSWER TO PART – I CASE SCENARIO BASED MCQS**

1. Option (b)
2. Option (c)
3. Option (b)
4. Option (c)
5. Option (c)
6. Option (c)
7. Option (b)
8. Option (d)
9. Option (b)
10. Option (c)
11. Option (b)
12. Option (c)
13. Option (c)
14. Option (b)
15. Option (a)

**ANSWERS OF PART – II DESCRIPTIVE QUESTIONS**

1. (a) Valuation of Startup under different scenarios:

(i) Best Case Scenario

|                        | Year 1       | Year 2       | Year 3       |               |
|------------------------|--------------|--------------|--------------|---------------|
| Revenue                | ₹ 100,00,000 | ₹ 120,00,000 | ₹ 144,00,000 |               |
| Expenses               | ₹ 80,00,000  | ₹ 92,40,000  | ₹ 108,00,000 |               |
| Cash Flow/<br>Earnings | ₹ 20,00,000  | ₹ 27,60,000  | ₹ 36,00,000  |               |
| Terminal Value         |              |              |              |               |
|                        |              |              |              | ₹ 3,60,00,000 |

|                  |             |             |             |               |
|------------------|-------------|-------------|-------------|---------------|
| PVF @ 20%        | 0.8333      | 0.6944      | 0.5787      | 0.5787        |
| PV               | ₹ 16,66,600 | ₹ 19,16,544 | ₹ 20,83,320 | ₹ 2,08,33,200 |
| Value of Startup |             |             |             | ₹ 2,64,99,664 |

(ii) Base Case Scenario

|                        | Year 1       | Year 2       | Year 3       |              |
|------------------------|--------------|--------------|--------------|--------------|
| Revenue                | ₹ 100,00,000 | ₹ 110,00,000 | ₹ 121,00,000 |              |
| Expenses               | ₹ 90,00,000  | ₹ 95,70,000  | ₹ 102,85,000 |              |
| Cash Flow/<br>Earnings | ₹ 10,00,000  | ₹ 14,30,000  | ₹ 18,15,000  |              |
| Terminal Value         |              |              |              | ₹ 181,50,000 |
| PVF @ 20%              | 0.8333       | 0.6944       | 0.5787       | 0.5787       |
| PV                     | ₹ 8,33,300   | ₹ 9,92,992   | ₹ 10,50,341  | ₹ 105,03,405 |
| Value of<br>Startup    |              |              |              | ₹ 133,80,038 |

(iii) Worst Case Scenario

|                        | Year 1       | Year 2       | Year 3       |             |
|------------------------|--------------|--------------|--------------|-------------|
| Revenue                | ₹ 100,00,000 | ₹ 102,00,000 | ₹ 104,04,000 |             |
| Expenses               | ₹ 95,00,000  | ₹ 98,94,000  | ₹ 101,95,920 |             |
| Cash Flow/<br>Earnings | ₹ 5,00,000   | ₹ 3,06,000   | ₹ 2,08,080   |             |
| Terminal Value         |              |              |              | ₹ 20,80,800 |
| PVF @ 20%              | 0.8333       | 0.6944       | 0.5787       | 0.5787      |
| PV                     | ₹ 4,16,650   | ₹ 2,12,486   | ₹ 1,20,416   | ₹ 12,04,159 |
| Value of Startup       |              |              |              | ₹ 19,53,711 |

Value of ABC Startup as per First Chicago Method

$$= 0.30 \times ₹ 2,64,99,664 + 0.60 \times ₹ 133,80,038 + 0.10 \times ₹ 19,53,711$$

$$= ₹ 79,49,899 + ₹ 80,28,023 + ₹ 1,95,371$$

$$= ₹ 1,61,73,293$$

(b) Some points to be kept in mind while preparing a Pitch Presentation are as follows:

(i) Introduction

(ii) Team

(iii) Problem

- (iv) Solution
- (v) Marketing/Sales
- (vi) Projections or Milestones
- (vii) Competition
- (viii) Business Model

Financial projections include three basic documents that make up a business's financial statements.

- **Income statement:** This estimate how much money the business will generate by projecting income and expenses. It will show:

- ❖ How much revenue did the business generate?
- ❖ How much did it cost to generate and support that revenue?
- ❖ How much did the business pay its employees?
- ❖ How much did it pay towards rent?

For your first year in business, you'll want to create a monthly income statement. For the second year, quarterly statements will suffice. For the following years, you'll just need an annual income statement.

- **Cash flow statement:** A projected cash flow statement will depict how much cash will be coming into the business and how much cash will be utilized. At the end of each period (e.g. monthly, quarterly, annually), one can tally it all up to show either the cash burn or the cash generated during the period and the cash balance remaining at the end of the period.
- **Balance sheet:** The balance sheet shows the business's overall finances including assets, liabilities and equity. Typically, one will create an annual balance sheet for one's financial projections. It shows:
  - ❖ How much cash is in the bank?
  - ❖ How much money does the company owe to suppliers?
  - ❖ How much money has been invested in the company?

2. (a) Profit After Tax (PAT) or Net Income = ₹ 8000 crores (1 – 0.30)  
= ₹ 5600 crores

Free Cash Flow to Equity (FCFE) = Net Income - Capital Expenditures + Depreciation -/+ Change in Net Working Capital + New Debt Issued - Debt Repayments + Net issue of Preference Shares – Preference Share Dividends

Free Cash Flow to Equity (FCFE) = ₹ 5600 crores - ₹ 20140 crore + ₹ 17100 crore - ₹ 1755.60 crore + ₹ 2062.108 crore = ₹ 2866.508 crore

Cost of Equity =  $R_f + \beta (R_m - R_f)$  or  $R_f + \beta$  Market Risk Premium  
= 9.50% + 0.1 x 3.10% = 9.81%

Value of Equity =  $\frac{FCFE(1+g)}{K_e - g} = \frac{2866.508 \text{ crore}(1.07)}{0.0981 - 0.07} = \frac{3067.1636 \text{ crore}}{0.0281}$   
= ₹ 109151.7295 crore

Value of one Equity Share =  $\frac{109151.7295 \text{ crore}}{380 \text{ crore}} = ₹ 287.24$

*Alternatively*, it can also be calculated by using per share basis as follows:

FCFE per share =  $\frac{FCFE}{\text{No. of Equity Shares}} = \frac{2866.508 \text{ crore}}{380 \text{ crore}} = ₹ 7.5434$

Value of per equity share =  $\frac{FCFE(1+g)}{K_e - g} = \frac{7.5434(1.07)}{0.0981 - 0.07} = \frac{8.0714}{0.0281}$   
= ₹ 287.24

- (b) Financial Analysis whether to set up the manufacturing units in India or not may be carried using NPV technique as follows:

I. Incremental Cash Outflows

|                                     | \$ Million |
|-------------------------------------|------------|
| Cost of Plant and Machinery         | 500.00     |
| Working Capital                     | 50.00      |
| Release of existing Working Capital | (15.00)    |
|                                     | 535.00     |

## II. Incremental Cash Inflow after Tax (CFAT)

### (1) Generated by investment in India for 5 years

|                                  | \$ Million |
|----------------------------------|------------|
| Sales Revenue (5 Million x \$80) | 400.00     |
| Less: Costs                      |            |
| Variable Cost (5 Million x \$20) | 100.00     |
| Fixed Cost                       | 30.00      |
| Depreciation (\$500Million/5)    | 100.00     |
| EBIT                             | 170.00     |
| Taxes@35%                        | 59.50      |
| EAT                              | 110.50     |
| Add: Depreciation                | 100.00     |
| CFAT (1-5 years)                 | 210.50     |

### (2) Cash flow at the end of the 5 years (Release of Working Capital) 35.00

### (3) Cash generation by exports (Opportunity Cost)

|                                          | \$ Million |
|------------------------------------------|------------|
| Sales Revenue (1.5 Million x \$80)       | 120.00     |
| Less: Variable Cost (1.5 Million x \$40) | 60.00      |
| Contribution before tax                  | 60.00      |
| Tax@35%                                  | 21.00      |
| CFAT (1-5 years)                         | 39.00      |

### (4) Additional CFAT attributable to Foreign Investment

|                                        | \$ Million |
|----------------------------------------|------------|
| Through setting up subsidiary in India | 210.50     |
| Through Exports in India               | 39.00      |
| CFAT (1-5 years)                       | 171.50     |

### III. Determination of NPV

| Year                  | CFAT (\$ Million) | PVF@12% | PV (\$ Million) |
|-----------------------|-------------------|---------|-----------------|
| 1-5                   | 171.50            | 3.6048  | 618.2232        |
| 5                     | 35                | 0.5674  | 19.8590         |
| Less: Initial Outflow |                   |         | 638.0822        |
|                       |                   |         | 535.0000        |
|                       |                   |         | 103.0822        |

Since NPV is positive the proposal should be accepted.

### 3. (a) (i) Portfolio Beta

$$0.20 \times 0.40 + 0.50 \times 0.50 + 0.30 \times 1.10 = 0.66$$

### (ii) Residual Variance

To determine Residual Variance first of all we shall compute the Systematic Risk as follows:

$$\beta_A^2 \times \sigma_M^2 = (0.40)^2(0.01) = 0.0016$$

$$\beta_B^2 \times \sigma_M^2 = (0.50)^2(0.01) = 0.0025$$

$$\beta_C^2 \times \sigma_M^2 = (1.10)^2(0.01) = 0.0121$$

Residual Variance

$$A \quad 0.015 - 0.0016 = 0.0134$$

$$B \quad 0.025 - 0.0025 = 0.0225$$

$$C \quad 0.100 - 0.0121 = 0.0879$$

### (iii) Portfolio variance using Sharpe Index Model

$$\text{Systematic Variance of Portfolio} = (0.10)^2 \times (0.66)^2 = 0.004356$$

$$\text{Unsystematic Variance of Portfolio} = 0.0134 \times (0.20)^2 + 0.0225 \times (0.50)^2 + 0.0879 \times (0.30)^2 = 0.014072$$

$$\text{Total Variance} = 0.004356 + 0.014072 = 0.018428$$

### (iv) Portfolio variance on the basis of Markowitz Theory

$$\begin{aligned}
&= (w_A \times w_A \times \sigma_A^2) + (w_A \times w_B \times \text{Cov}_{AB}) + (w_A \times w_C \times \text{Cov}_{AC}) + (w_B \times \\
&w_A \times \text{Cov}_{AB}) + (w_B \times w_B \times \sigma_B^2) + (w_B \times w_C \times \text{Cov}_{BC}) + (w_C \times \\
&w_A \times \text{Cov}_{CA}) + (w_C \times w_B \times \text{Cov}_{CB}) + (w_C \times w_C \times \sigma_C^2) \\
&= (0.20 \times 0.20 \times 0.015) + (0.20 \times 0.50 \times 0.030) + (0.20 \times 0.30 \\
&\times 0.020) + (0.20 \times 0.50 \times 0.030) + (0.50 \times 0.50 \times 0.025) + (0.50 \\
&\times 0.30 \times 0.040) + (0.30 \times 0.20 \times 0.020) + (0.30 \times 0.50 \times 0.040) \\
&+ (0.30 \times 0.30 \times 0.10) \\
&= 0.0006 + 0.0030 + 0.0012 + 0.0030 + 0.00625 + 0.0060 + \\
&0.0012 + 0.0060 + 0.0090 = 0.0363
\end{aligned}$$

(b) Various types of Interest rate risk faced by companies/ banks are as follows:

- (1) **Gap Exposure:** A gap or mismatch risk arises from holding assets and liabilities and off-balance sheet items with different principal amounts, maturity dates or re-pricing dates, thereby creating exposure to unexpected changes in the level of market interest rates. This exposure is more important in relation to banking business.
- (2) **Basis Risk:** Market interest rates of various instruments seldom change by the same degree during a given period of time. The risk that the interest rate of different assets, liabilities and off-balance sheet items may change in different magnitude is termed as basis risk. For example, while assets may be benchmarked to Fixed Rate of Interest, liabilities may be benchmarked to floating rate of interest. The degree of basis risk is fairly high in respect of banks that create composite assets out of composite liabilities.
- (3) **Embedded Option Risk:** Significant changes in market interest rates create another source of risk to banks' profitability by encouraging prepayment of cash credit/demand loans/term loans and exercise of call/put options on bonds/debentures and/or premature withdrawal of term deposits before their stated maturities.
- (4) **Yield Curve Risk:** The movements in yield curve are rather frequent when the economy moves through business cycles.

Thus, banks should evaluate the movement in yield curves and its impact on the portfolio values and income.

- (5) **Price Risk:** Price risk occurs when assets are sold before their stated maturities. In the financial market, bond prices and yields are inversely related. The price risk is closely associated with the trading book, which is created for making profit out of short-term movements in interest rates.

Banks which have an active trading book should, therefore, formulate policies to limit the portfolio size, holding period, duration, defeasance period, stop loss limits, marking to market, etc.

- (6) **Reinvestment Risk:** Uncertainty with regard to interest rate at which the future cash flows could be reinvested is called reinvestment risk. Any mismatches in cash flows would expose the banks to variations in NII as the market interest rates move in different directions.

**OR**

- (b) Some of the areas where the Blockchain can be applied are as follows:

- (i) **Financial Services:** Blockchain can be used to provide an automated trade lifecycle in terms of the transaction log of any transaction of asset or property - whether physical or digital such as laptops, smartphones, automobiles, real estate, etc. from one person to another.
- (ii) **Healthcare:** Blockchain provides secure sharing of data in healthcare industry by increasing the privacy, security, and interoperability of the data by eliminating the interference of third party and avoiding the overhead costs.
- (iii) **Government:** At the government front, there are instances where the technical decentralization is necessary but politically should be governed by governments like land registration, vehicle registration and management, e-voting etc. Blockchain improves the transparency and provides a better way to monitor and audit the transactions in these systems.

- (iv) **Travel Industry:** Blockchain can be applied in money transactions and in storing important documents like passports/other identification cards, reservations and managing travel insurance, loyalty, and rewards thus, changing the working of travel and hospitality industry.
- (v) **Economic Forecasts:** Blockchain makes possible the financial and economic forecasts based on decentralized prediction markets, decentralized voting, and stock trading, thus enabling the organizations to plan and shape their businesses.

4. (a) **Calculation of Income available for Distribution**

|                                                   | Units<br>(Lakh) | Per<br>Unit<br>(₹) | Total<br>(₹ In<br>lakh) |
|---------------------------------------------------|-----------------|--------------------|-------------------------|
| Income from April                                 | 300             | 0.0765             | 22.9500                 |
| Add: Dividend equalization<br>collected on issue  | 6               | 0.0765             | 0.4590                  |
|                                                   | 306             | 0.0765             | 23.4090                 |
| Add: Income from May                              |                 | 0.1125             | 34.4250                 |
|                                                   | 306             | 0.1890             | 57.8340                 |
| Less: Dividend equalization<br>paid on repurchase | 3               | 0.1890             | (0.5670)                |
|                                                   | 303             | 0.1890             | 57.2670                 |
| Add: Income from June                             |                 | 0.1500             | 45.4500                 |
|                                                   | 303             | 0.3390             | 102.7170                |
| Less: Dividend Paid                               |                 | 0.2373             | (71.9019)               |
|                                                   | 303             | 0.1017             | 30.8151                 |

**Calculation of Issue Price at the end of April**

|                                                   | ₹       |
|---------------------------------------------------|---------|
| Opening NAV                                       | 18.750  |
| Add: Entry Load 2% of ₹ 18.750                    | 0.375   |
|                                                   | 19.125  |
| Add: Dividend Equalization paid on Issue<br>Price | 0.0765  |
|                                                   | 19.2015 |

### Calculation of Repurchase Price at the end of May

|                                                | ₹       |
|------------------------------------------------|---------|
| Opening NAV                                    | 18.750  |
| Less: Exit Load 2% of ₹ 18.750                 | (0.375) |
|                                                | 18.375  |
| Add: Dividend Equalization paid on Issue Price | 0.1890  |
|                                                | 18.564  |

### Closing NAV

|                                            |          | ₹ (Lakh)    |
|--------------------------------------------|----------|-------------|
| Opening Net Asset Value (₹ 18.75 × 300)    |          | 5625.0000   |
| Portfolio Value Appreciation               |          | 425.4700    |
| Issue of Fresh Units (6 × 19.2015)         |          | 115.2090    |
| Income Received (22.950 + 34.425 + 45.450) |          | 102.8250    |
|                                            |          | 6268.504    |
| Less: Units repurchased (3 × 18.564)       | -55.692  |             |
| Income Distributed                         | -71.9019 | (-127.5939) |
| Closing Net Asset Value                    |          | 6140.9101   |
| Closing Units (300 + 6 – 3) lakh           |          | 303 lakh    |
| ∴ Closing NAV as on 30 <sup>th</sup> June  |          | ₹ 20.2670   |

(b) The arbitrageur can proceed as stated below to realize arbitrage gains.

(i) Buy ₹ from USD 10,000,000 At Mumbai  $48.30 \times 10,000,000$   
₹ 483,000,000

(ii) Convert these ₹ to GBP at London  $\left( \frac{₹ 483,000,000}{₹ 77.52} \right)$

GBP 6,230,650.155

(iii) Convert GBP to USD at New York  $GBP 6,230,650.155 \times 1.6231$  USD 10,112,968.26

There is net gain of USD 10,112,968.26 less USD 10,000,000 i.e. USD 112,968.26

**5. (a) Working Notes:**

**(i) Computation of Net Worth Per Share of SVL**

|                                                                                               |         |
|-----------------------------------------------------------------------------------------------|---------|
| Total Assets (Fixed assets + Current Assets) (₹ Crores)                                       | 2260    |
| Less: Liabilities (Current Liabilities + Borrowings) (₹ Crores)                               | 690     |
| Net Assets Value (₹ Crores)                                                                   | 1570    |
| Current Value of Land after growing for three years<br>@ 30% = $380 \times 1.2475$ (₹ Crores) | 474.05* |
| Less: Book Value (₹ Crores)                                                                   | 380.00  |
| Increase in the Value of land (₹ Crores)                                                      | 94.05   |
| Adjusted NAV (1570 + 94.05) (₹ Crores)                                                        | 1664.05 |
| No. Shares (Crores)                                                                           | 25      |
| Net Worth Per Share                                                                           | ₹ 66.56 |

\*Alternatively, this value can also be computed as ₹ 475 Crores.

**(ii) Computation of Net Worth Per Share of ICL**

Share Capital + Reserves and Surplus = ₹ 2600 Crore

Total Number of Shares = 50 Crore

Net Worth Per Share = ₹ 2600 Crore / 50 Crore = ₹ 52.00

**(iii) Earning Per Share (EPS)**

|               | ICL          | SVL         |
|---------------|--------------|-------------|
| PAT           | ₹ 1580 Crore | ₹ 500 Crore |
| No. of Shares | 50 Crore     | 25 Crore    |
| EPS           | ₹ 31.60      | ₹ 20.00     |

**(iv) Share price as per Dividend Growth Model**

|                                      | ICL                                     | SVL                                  |
|--------------------------------------|-----------------------------------------|--------------------------------------|
| Total Dividend                       | ₹ 470 Crore                             | ₹ 304.35 Crore                       |
| No. of Shares                        | 50 Crore                                | 25 Crore                             |
| Dividend Per Share (D <sub>0</sub> ) | ₹ 9.40                                  | ₹ 12.17                              |
| Expected Dividend (D <sub>1</sub> )  | ₹ 9.40 (1 + 0.18)<br>= ₹ 11.09          | ₹ 12.17 (1 + 0.15)<br>= ₹ 14.00      |
| Value of Per Share as Growth Model   | $\frac{11.09}{0.25-0.18}$<br>= ₹ 158.43 | $\frac{14.00}{0.20-0.15}$<br>= ₹ 280 |

**Calculation of Swap Ratio**

|                                          |               |                           |
|------------------------------------------|---------------|---------------------------|
| Net Worth Per Share                      | 1 : 1.28 i.e. | $1.28 \times 0.32$<br>25% |
| EPS                                      | 1 : 0.63 i.e. | $0.63 \times 0.19$<br>30% |
| Share price as per Dividend Growth Model | 1 : 1.77 i.e. | $1.77 \times 0.35$<br>20% |
| Market Price                             | 1 : 0.56 i.e. | $0.56 \times 0.14$<br>25% |
|                                          |               | Total <u>1.00</u>         |

Swap ratio is for every one share of SVL, to issue 1 share of ICL. Hence, total no. of shares to be issued 25 crores.

**(b) The characteristics of GDRs are as follows:**

- (i) Holders of GDRs participate in the economic benefits of being ordinary shareholders, though they do not have voting rights.
- (ii) GDRs are settled through CEDEL & Euro-clear international book entry systems.
- (iii) GDRs are listed on the Luxemburg stock exchange.
- (iv) Trading takes place between professional market makers on an OTC (over the counter) basis.

- (v) The instruments are freely traded.
- (vi) They are marketed globally without being confined to borders of any market or country as it can be traded in more than one currency.
- (vii) Investors earn fixed income by way of dividends which are paid in issuer currency converted into dollars by depository and paid to investors and hence exchange risk is with investor.
- (viii) As far as the case of liquidation of GDRs is concerned, an investor may get the GDR cancelled any time after a cooling period of 45 days. A non-resident holder of GDRs may ask the overseas bank (depository) to redeem (cancel) the GDRs. In that case overseas depository bank shall request the domestic custodians bank to cancel the GDR and to get the corresponding underlying shares released in favour of non-resident investor. The price of the ordinary shares of the issuing company prevailing in the Bombay Stock Exchange or the National Stock Exchange on the date of advice of redemption shall be taken as the cost of acquisition of the underlying ordinary share.

6. (a) Initial Margin =  $\mu + 3\sigma$

Where  $\mu$  = Daily Absolute Change

$\sigma$  = Standard Deviation

Accordingly

Initial Margin = ₹ 10,000 + ₹ 6,000 = ₹ 16,000

Maintenance margin = ₹ 16,000 x 0.75 = ₹ 12,000

| Day     | Changes in future Values (₹)     | Margin A/c (₹) | Call Money (₹) |
|---------|----------------------------------|----------------|----------------|
| 4/2/09  | -                                | 16000          | -              |
| 5/2/09  | 50 x (3294.40 - 3296.50) = -105  | 15895          | -              |
| 6/2/09  | 50 x (3230.40 - 3294.40) = -3200 | 12695          | -              |
| 7/2/09  | 50 x (3212.30 - 3230.40) = -905  | 16000          | 4210           |
| 10/2/09 | 50 x (3267.50 - 3212.30) = 2760  | 18760          | -              |
| 11/2/09 | 50 x (3263.80 - 3267.50) = -185  | 18575          | -              |

|         |                                         |       |      |
|---------|-----------------------------------------|-------|------|
| 12/2/09 | $50 \times (3292 - 3263.80) = 1410$     | 19985 | -    |
| 14/2/09 | $50 \times (3309.30 - 3292) = 865$      | 20850 | -    |
| 17/2/09 | $50 \times (3257.80 - 3309.30) = -2575$ | 18275 | -    |
| 18/2/09 | $50 \times (3102.60 - 3257.80) = -7760$ | 16000 | 5485 |

(b) To some extent this statement is correct. The advocates of technical analysis offer the following interrelated argument in their favour:

- (i) Under influence of crowd psychology trend persist for some time. Tools of technical analysis help in identifying these trends early and help in investment decision making.
- (ii) Shift in demand and supply are gradual rather than instantaneous. Technical analysis helps in detecting this shift rather early and hence provides clues to future price movements.
- (iii) Fundamental information about a company is observed and assimilated by the market over a period of time. Hence price movement tends to continue more or less in same direction till the information is fully assimilated in the stock price.

Detractors of technical analysis believe that it is an useless exercise; their arguments are as follows:

- (i) Most technical analysts are not able to offer a convincing explanation for the tools employed by them.
- (ii) Empirical evidence in support of random walk hypothesis casts its shadow over the usefulness of technical analysis.
- (v) By the time an uptrend and down trend may have been signalled by technical analysis it may already have taken place.
- (iv) Ultimately technical analysis must be a self-defeating proposition. With more and more people employing it, the value of such analysis tends to decline.